Discovering Ancient Egypt: An AI-Powered Guide

Ahmed Medhat, Ali Hamed, Ammar Nashaat, Hussien Osama, Youssef Sharaf, Ziad Tohamy

School of Information Technology and Computer Science

Nile University

Supervisor: Dr. Tamer Arafa

***Abstract*—With advancements in Large Language Models (LLMs) and multimodal, AI systems are increasingly capable of performing as knowledgeable, interactive guides in diverse applications. This paper surveys the potential of multimodal large language models (MLLMs), focusing on their use as an AI-enhanced educational tool tailored for learning about ancient Egypt. Leveraging LLAMA 3.2, this paper explores a chatbot designed to serve as an AI-powered guide for tourists and learners, capable of answering questions, providing historical insights, and recognizing Egyptian monuments through image retrieval and contextual description. By synthesizing insights from recent innovations in Retrieval-Augmented Generation (RAG) and fine-tuning methodologies, we analyze methods for creating accurate, engaging, and culturally enriched responses in conversational interfaces. This survey not only presents key methodologies for developing educational AI applications but also addresses challenges and future directions, underscoring the role of MLLMs in transforming historical education for tourists and scholars alike.**

***Index Terms*—Llama-3.2, Chatbot, Ancient Egypt, MLLM, RAG, Tourism**

I. Introduction

The rapid evolution of artificial intelligence (AI) and large language models (LLMs) has brought transformative changes to various domains, including cultural heritage and tourism. However, several challenges persist in applying these technologies effectively. A key issue in designing AI systems for historical and cultural contexts is achieving contextual accuracy and adaptivity, as many LLMs face limitations in understanding domain-specific nuances and handling multimodal data (Ab dallah & El-Khawaga, 2022). In this project, LLAMA 3.2, a state-of-the-art multimodal large language model (MLLM), is deployed to tackle these challenges by integrating advanced capabilities for both text and visual inputs, offering a unique approach to exploring ancient Egyptian heritage. Existing LLMs like LLAMA 2 have demonstrated the ability to process vast amounts of data, yet they often require significant fine-tuning to adapt to specialized tasks, such as identifying historical monuments or interpreting ancient artifacts (Fuchs et al., 2019; Li & Wang, 2023). LLAMA 3.2 builds upon this foundation by incorporating data from diverse sources, including structured historical databases, visual datasets of Egyptian artifacts, and textual descriptions, allowing for enhanced contextual and multimodal understanding (Cui et al., 2024). Despite the availability of extensive training data, achieving real-time relevance and accuracy in tourist-facing applications remains a significant challenge, particularly when handling user-generated queries or images (Ahmad & Hasan, 2023). The project also addresses limitations in traditional text-based LLM interactions by leveraging the Retriever Augmented Generation (RAG) framework, which enables dynamic retrieval of external information. This approach resolves issues of outdated or incomplete responses by accessing up-to-date, contextually relevant data (Kumar & Sharma, 2022; Vakayil et al., 2024). RAG systems rely heavily on vector databases, which use semantic embeddings to store and retrieve information efficiently, yet ensuring the quality and coverage of the underlying data remains a concern (Maged et al., 2024). Multimodal integration, another cornerstone of this project, provides the chatbot with image recognition capabilities to analyze and interpret visual inputs from users, such as photographs of monuments or artifacts. While this enhances interactivity and immersion, multimodal AI systems often struggle with aligning visual and textual data in a meaningful way (Fuchs et al., 2019; Omar & Farag, 2023). LLAMA 3.2 overcomes these limitations by fine-tuning its multimodal capabilities on a curated dataset specific to ancient Egyptian heritage, ensuring relevance and accuracy in its responses. Moreover, while natural language processing (NLP) technologies have improved significantly, challenges such as cultural bias and the need for multilingual support remain prevalent in heritage applications (Cheng & Wu, 2022; Smith & Zhang, 2020). These issues are mitigated in this project by employing adaptive NLP methods tailored to the linguistic and cultural context of ancient Egypt, ensuring inclusivity and accessibility for a diverse audience of tourists. By addressing these challenges through advanced AI technologies, this project not only enhances the accessibility of Egyptian heritage but also demonstrates the potential of AI in creating personalized and engaging educational experiences. The integration of cutting-edge frameworks, such as RAG and multimodal processing, positions this chatbot as a valuable tool for both tourism and cultural preservation (Williams & Dawson, 2022).

II. **DATASETS**

The development of an effective AI-powered chatbot, such as the one outlined in this project, relies heavily on the quality and comprehensiveness of its dataset. By employing advanced web scraping techniques and integrating reliable secondary sources, the chatbot ensures its responses are accurate, contextually relevant, and authoritative for users exploring ancient Egyptian history. Data Collection through Web Scraping is to ensure the chatbot provides accurate and reliable content, image and text web scraping techniques are employed to gather information and data from authoritative sources. Certified websites, such as the official websites of Egyptian museums and academic research publications, are scraped for images, descriptions, and historical data related to monuments, artifacts, and ancient Egyptian culture. The role of web scraping in this project is to continuously gather up-to-date, validated content that can be incorporated into the chatbot's knowledge base. This process enables the chatbot to present images of artifacts or monuments alongside detailed textual descriptions, ensuring users receive reliable, high-quality information. The importance of image and text web scraping lies in enriching the chatbot's dataset, ensuring it remains current, accurate, and comprehensive. It enhances the chatbot's ability to present users with verified content, further increasing the tool's trustworthiness and educational value.

A. ***Primary Dataset:*** *Egyptian Museum Website*

To ensure the chatbot provides reliable information, data was web-scraped from the official website of the Egyptian Museum. This dataset includes:

- **Images:** High-quality labeled images of artifacts, monuments, and exhibits.
- **Text Data:** Detailed descriptions of artifacts, highlighting their historical significance, cultural context, and archaeological findings

As shown in Fig.1, The Egyptian Museum's website was chosen as a primary source due to its authority and accuracy as shown in the following figure in representing Egypt's historical heritage. By incorporating this web-scraped data, the chatbot delivers an authentic and trustworthy user experience.

B. ***Supplementary Dataset:*** *Historical Dictionary of Ancient Egypt*

To further enrich the chatbot's knowledge base, text from the second edition of the Historical Dictionary of Ancient Egypt by Morris L. Bierbrier was integrated.
This authoritative source provides:

•**Ancient Egyptian Dynasties and Families:** Comprehensive information on ruling dynasties and notable figures.

• **Key Historical Events:** Detailed accounts of significant events in ancient Egyptian history.

• **Cultural Insights:** Explanations of cultural practices, religious beliefs, and societal norms.

The integration of this supplementary text ensures the chatbot can address complex historical topics, offering in-depth explanations about specific dynasties, events, or cultural practices.

By leveraging both web scraping and reliable secondary sources, the chatbot delivers an engaging and informative user experience, ensuring it remains a valuable educational tool for exploring Egypt's rich heritage.

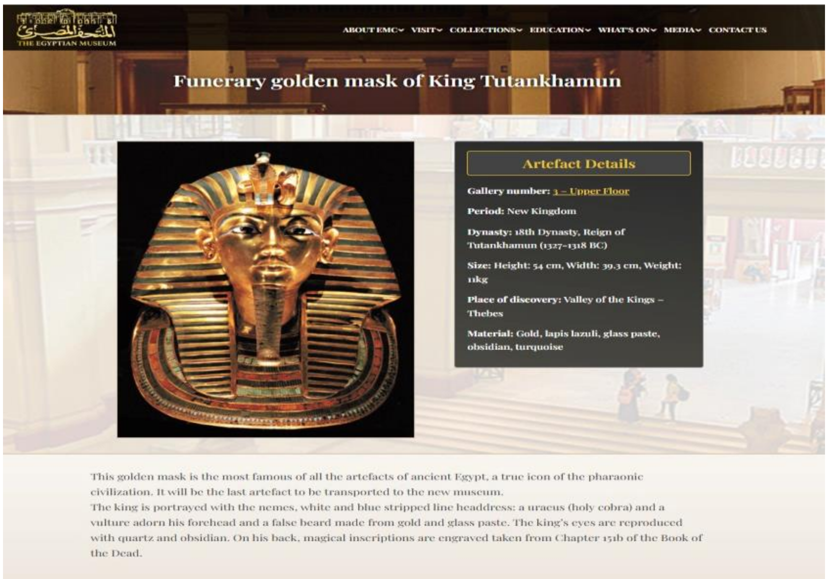

Fig. 1. The funerary golden mask of Tutankhamun. One of the pages we web-scraped from the primary dataset obtained from the Egyptian Museum website features this iconic artifact. The page includes high-resolution images and detailed textual descriptions that highlight the mask's historical significance, cultural context, and craftsmanship. This dataset serves as a cornerstone for the chatbot, ensuring accurate and engaging educational content about ancient Egyptian heritage.

III. Literature Review

The application of AI and large language models (LLMs) in cultural heritage and historical document recognition has seen significant advancements, creating new possibilities for enhancing the tourist experience. Dutta and Banerjee (2020) provide a comprehensive review of deep learning methods for historical document analysis, demonstrating the field's transition from traditional approaches to advanced AI-powered techniques. Their work highlights the evolution of these technologies in recognizing and interpreting historical texts, which aligns with the objective of the AI-powered chatbot to provide detailed information about Egyptian monuments through modern algorithms (Dutta and Banerjee, 2020).

Monuments, as vital cultural artifacts, pose distinct challenges in recognition and classification, particularly when catering to diverse visitor interests. Hassan et al. (2023) examine various methodologies for monument recognition across different cultural contexts, including Egypt, emphasizing the complexities inherent in recognizing and interpreting unique artifacts. Their survey underscores the importance of sophisticated recognition models, such as those employed in this project, to accurately process monument images and deliver relevant information to tourists (Hassan et al., 2023). In the realm of Egyptian hieroglyphics, Barucci et al. (2021) explore the capabilities of convolutional neural networks (CNNs) for the classification of ancient symbols, creating Glyphnet, a CNN specifically for hieroglyph classification. This work demonstrates the potential of AI-driven models to process and classify historical artifacts accurately, an approach that can similarly be applied in this project to facilitate tourist engagement with Egyptian hieroglyphs and other artifacts through image-based recognition (Barucci et al., 2021). Recent developments in multimodal LLMs have shown remarkable capabilities in processing and understanding extended image sequences, which is crucial for providing context to tourists viewing sequential images or artifacts. Ye et al. (2024) introduce mPLUG-Owl3, an advanced model for long image-sequence understanding, achieving state-of-the-art performance by integrating both vision and language. This research supports the project's goal of using a multimodal approach with LLAMA 3.2, enabling tourists to gain a layered understanding of historical contexts via combined text and image information (Ye et al., 2024). The use of retrieval-augmented generation (RAG) systems for image retrieval is also pertinent to this project. Bag et al. (2024) present a novel approach for image extraction in RAG-powered systems, which improves traditional methods by combining language processing with image recognition. Their methodology informs the RAG system used in this project, which aims to dynamically retrieve contextual information on Egyptian artifacts and sites, providing tourists with accurate and detailed responses to their inquiries (Bag et al., 2024). Joshi et al. (2024) further emphasize the efficacy of multimodal RAG systems in enhancing retrieval accuracy. By implementing a robust pipeline that processes various document types, including text, images, and tables, Joshi et al. demonstrate how multimodal integration can improve AI-driven educational tools. Their findings highlight the advantages of a multimodal approach for this project, which strives to offer tourists a comprehensive educational experience by integrating text and images (Joshi et al., 2024). Finally, in the context of Arabic natural language processing, Maged et al. (2024) examine LLMs in answering questions about Egyptian history. Their findings reveal both the challenges and opportunities in applying LLMs to domain-specific content, such as historical and cultural information. This research supports the chatbot's goal to deliver accurate and engaging information to tourists, particularly in answering questions related to Egyptian history and monuments, addressing both language processing needs and the cultural depth required for effective engagement (Maged et al., 2024).

IV. TECHNOLOGIES AND METHODS

The AI-enhanced chatbot for Learning about Ancient Egypt integrates a range of cutting-edge technologies and methodologies, each playing a pivotal role in ensuring the chatbot's effectiveness in delivering an engaging and informative experience for users. This section outlines the core technologies employed in the project, their functionalities, and their importance in achieving the goal of providing an interactive and immersive learning experience for tourists interested in ancient Egyptian history.

A. Natural Language Processing (NLP)

Natural Language Processing (NLP) serves as the fundamental technology powering the chatbot's ability to understand and interact with users in a human-like manner. NLP involves the application of computational techniques to process and analyze human language, enabling machines to interpret, generate, and respond to user inputs effectively (Abdallah & El-Khawaga, 2022; Chatterjee et al., 2021). In the context of this project, NLP allows the chatbot to understand questions, interpret historical queries, and generate coherent and contextually appropriate responses.

The importance of NLP lies in its ability to make the interaction between the user and the chatbot seamless and natural. As tourists inquire about various historical artifacts, monuments, and aspects of ancient Egyptian civilization, the chatbot can respond with precise, contextually relevant information (Omar & Farag, 2023). The chatbot's NLP capabilities also enable it to discern the intent behind user questions, ensuring it provides accurate answers, whether they involve specific facts, broad explanations, or detailed descriptions of monuments and artifacts (Ahmad & Hasan, 2023). Without NLP, the chatbot would struggle to comprehend the vast range of user queries and provide relevant answers, severely limiting its educational potential.

B. Multi-Large Language Models

The AI-powered chatbot utilizes LLAMA 3.2 11B, a state-of-the-art Multi-Modal Large Language Model (MLLM), to generate contextually relevant, coherent, and meaningful responses for users. LLAMA 3.2 11B is specifically designed to handle complex language tasks, leveraging a vast corpus of pre-existing data to understand nuances in language and context. This makes it a critical tool for addressing user queries related to ancient Egypt, such as identifying monuments, explaining historical events, and describing artifacts, while maintaining an intuitive and conversational approach (Abdallah & El-Khawaga, 2022; Fuchs et al., 2019).

The model's advanced architecture enables it to deliver personalized responses that adapt to individual users' preferences, knowledge levels, and learning styles. For instance, the chatbot can adjust the complexity of its answers based on a user's familiarity with ancient Egyptian history, ensuring accessibility for both beginners and experts alike (Williams & Dawson, 2022). This dynamic interaction makes the chatbot a valuable educational tool, blending accuracy with conversational engagement. In addition to its text-based capabilities, LLAMA 3.2 11B integrates multimodal functionalities, enabling it to process both textual and visual inputs. This multimodal approach enhances the chatbot's ability to deliver an immersive learning experience by analyzing and responding to user-submitted images, such as photographs of artifacts or monuments (Ahmad & Hasan, 2023). The inclusion of visual aids complements textual explanations, allowing users to gain deeper insights into Egypt's cultural and historical heritage. By leveraging MLLMs, the chatbot bridges the gap between textual and visual information, creating a comprehensive, multimedia-rich environment for users. For example, when a tourist inquiry about a specific monument or artifact, the chatbot can retrieve and display an image of the item, overlayed with detailed historical and cultural information (Omar & Farag, 2023). This functionality enhances interactivity and caters to diverse learning preferences, appealing to users who benefit from visual representations as well as those who prefer textual details. The integration of multimodal capabilities addresses a significant gap in traditional AI systems, which often struggle to align visual and textual data meaningfully (Fuchs et al., 2019). By adopting LLAMA 3.2 11B, this project offers tourists a richer and more interactive way to engage with ancient Egyptian heritage, showcasing the transformative potential of AI technologies in cultural tourism and education.

C. Retrieval-Augmented Generation (RAG)

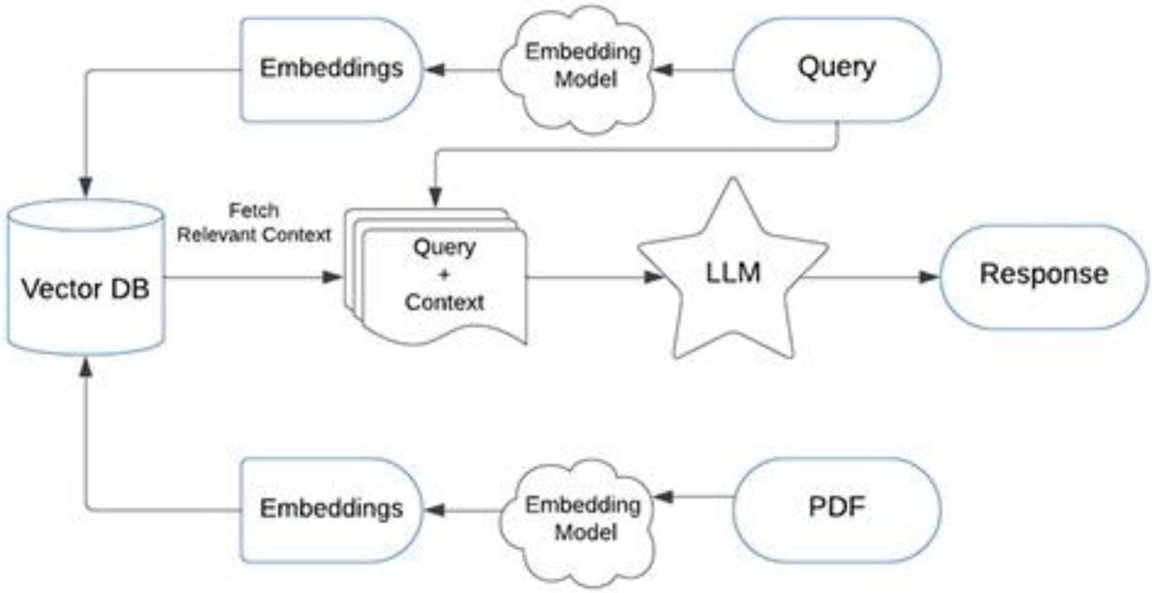

Fig. 2. Workflow Diagram of a Retrieval-Augmented Generation (RAG) System (Gupta et al., 2023).

Retrieval-Augmented Generation (RAG) is an advanced methodology used to enhance the chatbot's information retrieval process. By combining the power of a language model with a retrieval system, RAG enables the chatbot to access and incorporate relevant information from external sources in real-time (Lewis et al., 2020; Izacard & Grave, 2021). This ensures that the chatbot can provide highly accurate, up-to-date responses by pulling in information from a wide range of sources, including certified websites, academic papers, and authoritative repositories (Karpukhin et al., 2020). The role of RAG in this project is to enhance the chatbot's ability to provide comprehensive and contextually rich responses. When a user asks a question, the RAG system retrieves relevant data from an external source, such as the Egyptian Museum's official website or a scholarly article about Egyptian artifacts, and then the chatbot generates a response using that retrieved information. This system ensures that the chatbot is not limited to its pre-programmed knowledge base and can continuously integrate new information, keeping the content fresh and accurate (Gupta et al., 2023).

As shown in Fig. 2, RAG is critical for ensuring the chatbot remains accurate, responsive, and capable of handling a wide range of queries. Given the scope of ancient Egyptian history, which spans millennia, the retrieval system ensures that after receiving the query from the user it gets converted into an embedded query vector preserving the semantics using an embedding model then it retrieves the top K-relevant content from the vector database by computing similarity between the query embedding and the content embedding in the database. after that, it Passes the retrieved content and query as a prompt to an MLLM and finally, the MLLM gives the required response.

D. Vector Databases

Vector databases play a crucial role in the efficient storage, retrieval, and organization of data within the chatbot. These databases store data in the form of vector embeddings, which represent text, images, or other data in high-dimensional space (Chandrasekaran et al., 2022; Johnson et al., 2017). By using vector embeddings, the chatbot can quickly and accurately retrieve the most relevant information based on user queries, even when the data comes from large, complex datasets (Wu et al., 2023). The role of vector databases in this project is to optimize the retrieval of information, enabling the chatbot to respond to user queries in real-time. For instance, when a tourist asks about a specific monument, the chatbot uses vector embeddings to search through its vast knowledge base and retrieve the most relevant historical data or visual content. This ensures that users receive immediate, contextually accurate answers, enhancing the overall user experience (Babenko & Lempitsky, 2016). The importance of vector databases lies in their ability to facilitate fast, efficient searches through massive amounts of data. This is essential for maintaining the chatbot's responsiveness, especially when users pose complex or varied queries. Without vector databases, the chatbot would struggle to quickly locate and retrieve the vast amount of historical information required for accurate responses.

V. COMPARATIVE STUDY

The advent of advanced AI technologies has transformed the way we process and understand complex information across various domains. Among these innovations, Multimodal Large Language Models (MLLMs) and Retrieval-Augmented Generation (RAG) systems have emerged as powerful tools for handling diverse data types, including text, images, and multimodal content. These systems have proven especially effective in applications requiring deep contextual understanding and precise retrieval of information from large datasets (Cho et al., 2024).

Building on this foundation, frameworks such as M3DOCRAG and instruction-tuned multimodal models like Qwen/Qwen2-VL-2B-Instruct and Qwen/Qwen2-VL-7B Instruct represent significant advancements in multimodal understanding. While these systems share foundational concepts, their implementations and objectives reveal distinct approaches to handling complex multimodal datasets. Notably, M3DOCRAG is employed as the retrieval model (ColPali) for both Qwen2-VL models, with a higher version (vidore/colpali v1.2) used for the Qwen/Qwen2-VL-7B-Instruct model (Cho et al., 2024).

In this survey, we compare our approach to that of Cho et al. (2024), analyzing the similarities and differences in how M3DOCRAG and our AI-enhanced chatbot leverage multi-modal retrieval and RAG for their respective objectives. While M3DOCRAG focuses on multi-page, multi-document understanding, our work explores the integration of LLAMA 3.2 11B, multimodal functionalities, RAG, and vector databases to create an engaging, interactive platform tailored for learning about ancient Egypt.

By utilizing data web-scraped from the Egyptian Museum website, organized into a PDF with images and descriptions, this chatbot bridges the gap between historical artifacts and digital accessibility. Through this comparative study, we aim to highlight the potential of multimodal AI systems in revolutionizing cultural heritage education and tourism. By drawing parallels to the methodologies and findings of M3DOCRAG, this survey contributes to the growing body of literature on leveraging MLLMs and RAG for specialized applications, emphasizing the role of multimodal retrieval and interactive technologies in enhancing user engagement and learning outcomes.

A. Implementation

1) *Overview:* M3DOCRAG (Multi-Modal Multi-Document Retrieval-Augmented Generation) is a state-of-the-art framework designed to address challenges in multi-page, multi-document understanding. The model leverages the synergy of multi-modal retrieval and retrieval-augmented generation (RAG) to process and comprehend both text and images from complex datasets. It excels in extracting relevant information from multiple pages of documents, making it particularly effective for tasks requiring in-depth document-level understanding and multimodal context alignment (Cho et al., 2024).

2) Model Architecture: The M3DOCRAG framework is composed of several interconnected modules, each optimized for a specific task as shown in Fig. 3 (Cho et al., 2024).

Also, in Fig. 3 comparison of multi-modal document understanding pipelines. Previous works focus on (a) single-page DocVQA that cannot handle many long documents or (b) text-based RAG that ignores visual information. Our (c) M3DOCRAG framework retrieves relevant documents and answers questions using multimodal retrieval and MLM components so that it can efficiently handle many long documents while preserving visual information. M3DOCRAG Framework: M3DOCRAG is designed to address challenges in multi-page, multi-document understanding. It emphasizes aligning textual and visual modalities for tasks requiring document-level comprehension. The framework integrates ColPali for document embedding, a vector database for efficient retrieval, and multi-modal large language models (MLLMs) like Qwen2-VL for question answering. Its modular architecture excels in structured and unstructured data processing (Cho et al., 2024).

As shown in the bellow Fig.4, their M3DOCRAG framework consists of three stages: (1) document embedding, (2) page retrieval, and (3) question answering. In (1) document embedding, we extract visual embedding (with ColPali) to represent each page of all PDF documents. In (2) page retrieval, we retrieve the top-K pages of high relevance (MaxSim scores) with text queries. In an open-domain setting, we create approximate page indices for a faster search. In (3) question answering, we conduct visual question answering with multi-modal LM (e.g. Qwen2-VL) to obtain the final answer.

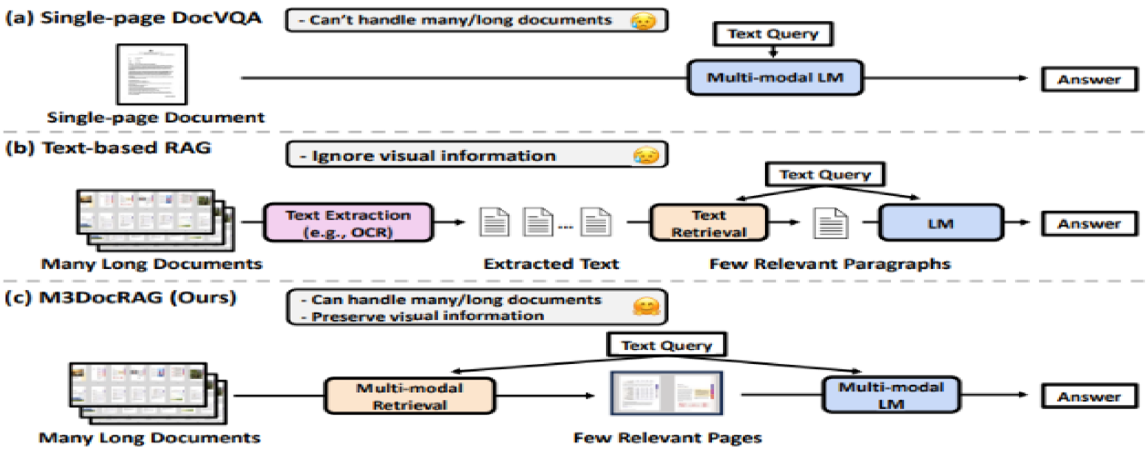

Fig. 3, Comparison of Document Question-Answering Approaches (Cho et al., 2024): Single-page DocVQA, Text-based RAG, and M3DocRAG

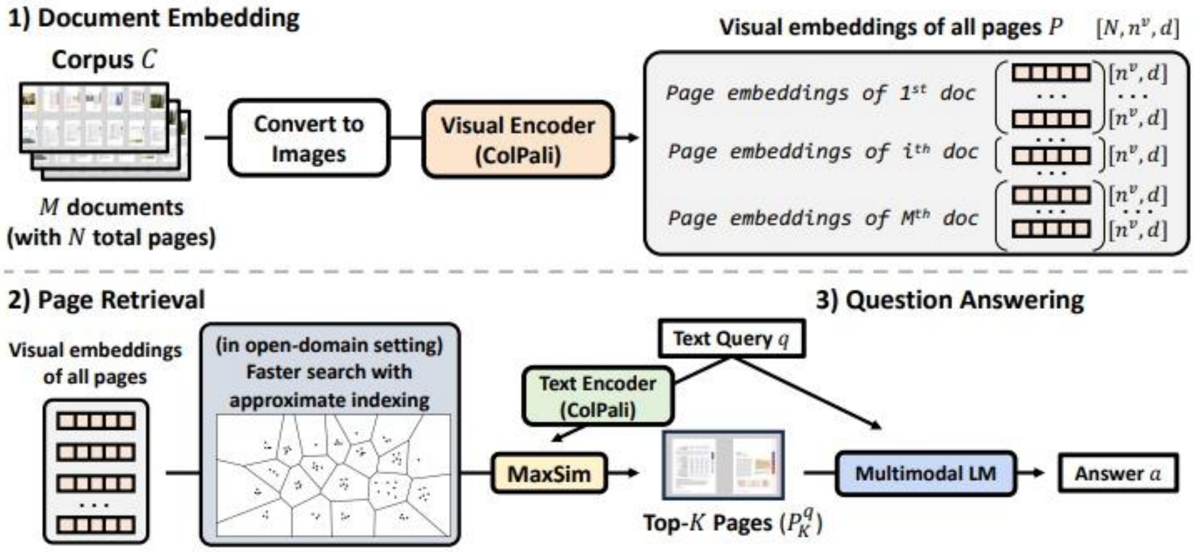

Fig. 4, Pipeline for Multi-modal Document Retrieval and Question Answering Using Visual and Text Embeddings (Cho et al., 2024)

The key architecture components include:

Input Preprocessing Module

- **Textual Input**: Text is extracted from multipage documents using optical character recognition (OCR) for scanned documents or direct text parsing for digital formats. The text is tokenized and processed to embeddings for semantic analysis.
- **Visual Input:** Images or diagrams present within the documents are processed using a vision model to extract relevant visual features. These features are represented as high-dimensional embeddings that align with textual data.

1. Embedding Module

- **Unified Embeddings:** M3DOCRAG employs pre-trained embedding models to convert both textual and visual data into a shared embedding space. This allows for seamless multi-modal alignment, enabling the system to process and compare information across modalities.
- **Page-Level Embeddings:** Each page in a document is represented as a combination of textual and visual embeddings, ensuring that all information within the page is captured.

2. Retriever

The retriever is responsible for identifying the most relevant sections from the multi-document dataset. It:

- Convert user queries into query embeddings.
- Computes similarity scores between the query embed dings and document embeddings stored in a vector database.
- Retrieves the top-k most relevant pages or sections from the database for further processing.

3. Multi-Modal RAG (Retrieval-Augmented Generation) Module

Once relevant sections are retrieved, the RAG module integrates this data with the user query to generate a coherent response. The process involves:

- Feeding the retrieved content and query into a pre-trained multi-modal large language model (MLLM).
- Generating a response that combines information from both textual and visual sources.

Output Generation

The output is presented as a unified response that combines text and optional visual references. For example, a response to a query about a specific artifact might include both a textual explanation and an annotated image.

Qwen/Qwen2-VL Models: The Qwen2-VL series represents cutting-edge instruction-following multimodal language models optimized for retrieval and generation tasks. The 2B-Instruct model utilizes ColPali for retrieval, while the 7B-Instruct model incorporates vidore/colpali-v1.2, offering enhanced retrieval capabilities. Both models ensure seamless integration of text and visual data, providing robust multimodal interactions for various applications (Qwen Team, 2024).

Key Model Differences:

- ***Retrieval Model Versions***

A) Qwen2-VL-2B relies on the standard ColPali model for embedding and retrieval.

B) Qwen2-VL-7B employs vidore/colpali-v1.2, which provides superior retrieval accuracy and multimodal alignment (Cho et al., 2024).

- ***Model Size and Capability***

A) Qwen2-VL-2B is smaller, focusing on lightweight, memory-efficient operations suitable for low-resource environments.

B) Qwen2-VL-7B offers a more nuanced understanding due to its larger model size, ideal for precision-critical applications (Qwen Team, 2024).

- ***Multimodal Interaction***

Both models leverage qwen-vl-utils for text-image integration, but Qwen2-VL-7B benefits from enhanced embeddings and processing capabilities.

In other architecture, the VisRAG (Vision-based Retrieval Augmented Generation) framework addresses the challenge of integrating multi-modal retrieval and generation within a unified system (Zhu et al., 2024). Its primary objective is to enhance the processing and generation of information from documents that contain both visual and textual elements by leveraging pre-trained vision language models. The core contributions of the framework include the proposal of a unified architecture that seamlessly combines retrieval and generation, leading to significant improvements in benchmarks related to multi-modal document understanding and response generation. The system design utilizes advanced pre-trained models to extract and align visual and textual features while incorporating a robust retrieval mechanism to ensure that contextually relevant information is provided for downstream generation tasks. The effectiveness of VisRAG has been validated on multi-modal datasets, where it has demonstrated superior performance in both retrieving and synthesizing information compared to baseline models (Zhu et al., 2024).

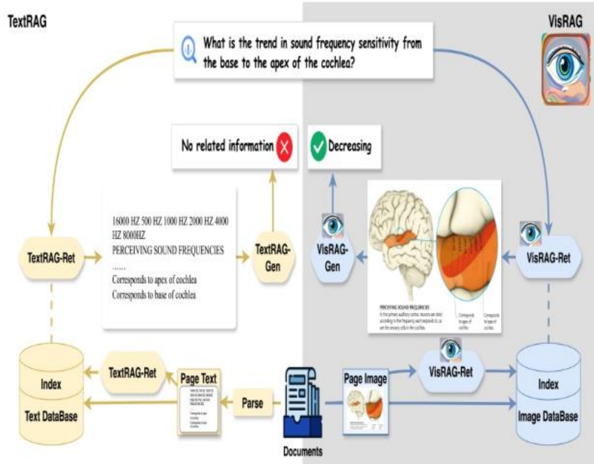

Fig. 5

As shown in Fig. 5, TextRAG (left) vs. VisRAG (right). Traditional text-based RAG (TextRAG) relies on parsed texts for retrieval and generation, losing visual information in multi-modal documents. Our vision-based RAG (VisRAG) employs a VLM-based retriever and generator to directly process the image of the document page, thus preserving all the information on the original page (Zhu et al., 2024).

The VisRAG framework is designed to enable vision-based retrieval-augmented generation for multi-modal documents. It focuses on both retrieval and generation capabilities within a cohesive system, with the aim of enhancing document comprehension and response generation by leveraging both visual and textual information.

1. **Objective**

VisRAG aims to facilitate efficient vision-based retrieval of augmented generation in tasks involving multi-modal documents. The framework integrates both retrieval and generation functionalities into a unified system, allowing for improved understanding and response generation using contextual information from both text and visual elements.

2. **Model Architecture**

a) Retrieval Model:

- **Feature Extraction and Alignment:**

  The retrieval component utilizes the openbmb/VisRAG-Ret model for extracting and aligning features from both visual and textual content.

- **Embedding Space:**

The model operates in a vision language embedding space, enabling a unified representation of visual and textual data.

- **Similarity Computation:**
  Features extracted are processed through a weighted mean pooling mechanism to generate embeddings, which are then normalized to allow efficient similarity computation for retrieval.

b) Generation Model:

- **Text Generation:** The generation component employs the openbmb/MiniCPM-V-2 model for text generation, ensuring the synthesized output is coherent and contextually relevant.
- **Contextual Integration:** The model integrates the multi-modal context retrieved by the retrieval mechanism into the generated responses, enhancing their relevance and richness.
- **Optimization:** The model is fine-tuned for generating contextually enriched, coherent responses based on the retrieved multi-modal information.

3. RAG Type

VisRAG implements a tightly integrated retrieval augmented generation (RAG) mechanism. It leverages document-level multi-modal embeddings to provide comprehensive contextual understanding, allowing for the generation of informed and contextually relevant outputs based on both visual and textual information. the VisRAG system primarily aligns with a dense retrieval-augmented generation (RAG) type, which is evident from its architecture and methodology. Specifically, the use of vision-language embedding spaces, along with pre-trained models and normalized feature vectors, strongly indicates a dense retrieval approach. In this system, embeddings are generated for both visual and textual data, which are then compared directly for similarity in a continuous vector space. This direct comparison of embeddings in the continuous space is a key characteristic of dense retrieval systems, distinguishing them from sparse retrieval methods that rely on keyword-based matching.

Finally, DAE AN AI POWERED SYSTEM

Now we are going to discuss the implementation of our model.

1. System Overview

Our system is a Multimodal Retrieval-Augmented Generation (RAG) framework that integrates text and image modalities to retrieve and generate relevant information. The implementation is designed to provide an interactive and intelligent interface for exploring artifacts and historical documents. The following subsections outline the key components of the system.

2. Data Preprocessing

We utilized a PDF processing pipeline to extract both text and image data from documents. Key steps included:

Text Extraction: Text was extracted from PDF documents using PyPDF2. Each page was processed to produce text chunks.
Image Extraction: Images were extracted from PDFs using a custom script leveraging PyPDF2's XObject resources. Extracted images were saved in an organized format for further processing.
Metadata Linking: Metadata was created to link each text chunk with its corresponding page and images, enabling precise alignment between textual and visual data.

3. Embedding and Storage

The extracted text and image data were embedded into a shared vector space using a combination of models:

Text Embeddings: Text chunks were encoded using the all-MiniLM-L6-v2 model from SentenceTransformers.
Image Embeddings: Images were processed using the CLIP (Contrastive Language-Image Pretraining) model (openai/clip-vit-base-patch32), which generates embeddings in a shared latent space for both text and images.
Projection Layer: A linear projection layer was implemented to map image embeddings to the same dimensionality as text embeddings for consistent storage.
The embeddings, along with their metadata, were stored in ChromaDB, a high-performance vector database optimized for retrieval tasks.

4. Retrieval Module

To retrieve relevant information based on user queries, we implemented a query mechanism that handles both text and image inputs:

Text Query: Text inputs are encoded into embeddings and compared with stored embeddings in ChromaDB using cosine similarity.
Image Query: For image-based queries, the CLIP model generates embeddings for the given image. These embeddings are then used to perform similarity searches in ChromaDB.
Top results from the database are returned, including both textual and visual data, providing a multimodal response.

5. User Interface

Text-based Search: Users can input text queries to retrieve relevant information.
Image-based Search: Users can upload images, and the system retrieves relevant text and images associated with the uploaded image.
Visual Display: Retrieved text is displayed alongside corresponding images, offering a comprehensive multimodal experience.

Challenges and Solutions

Data Alignment: Linking text chunks with corresponding images required robust metadata creation. This was addressed by developing a custom extraction and alignment pipeline.

Embedding Dimensionality: Image and text embeddings had mismatched dimensions. A projection layer resolved this issue, ensuring consistent representation in the vector space.

Efficient Retrieval: Using ChromaDB enabled rapid similarity searches, even with a large corpus of embeddings.

Example Workflow

A user provides a query: "*What do you know about Coffins_of_Padiamun_7?*"

The system encodes the query and searches ChromaDB for the most relevant text and image embeddings.

Retrieved results include textual descriptions of the artifact, accompanied by images extracted from the source document.

Results are displayed in a visually organized manner, enhancing user understanding and engagement.

Fig. 6 User Interface

B. **Comparison**

The M3DOCRAG and Discovering Ancient Egypt: An AI-Powered Guide both utilize multi-modal retrieval systems to process and integrate textual and visual data. However, their implementation and objectives differ significantly.

1) **AI-Enhanced Chatbot's Approach:**

The AI-Enhanced Chatbot is designed to serve as an interactive, AI-powered tourism assistant that bridges the gap between visitors and Ancient Egyptian cultural heritage. Its primary objective is to provide users with an engaging, informative experience by combining both text and visual data to answer queries related to Ancient Egyptian history. The system utilizes a retrieval model powered by a vector database that stores vector embeddings of text and images, which are then retrieved based on user queries to ensure real-time, contextually relevant responses. The retrieval process is optimized for efficiency, offering fast and user-friendly access to relevant content. The chatbot's generation model, powered by the *Llama 3.2 11B multi-modal model*, integrates the retrieved text and images into natural language responses tailored to the user's needs. This approach implements an interactive retrieval-augmented generation (RAG) system, focusing on contextual relevance and ease of use. The data pipeline processes user queries by retrieving the top-k nearest neighbors from the vector database, passing this content to the generation model, and ensuring seamless integration of both text and images in the responses. This combination of advanced retrieval and generation techniques ensures a rich, informative, and user-centered experience for exploring Ancient Egyptian heritage.

2) **M3DOCRAG's Approach:**

M3DOCRAG employs an advanced **multi-modal retrieval mechanism** that combines information from images and text within multi-page, multi-document contexts. It excels in aligning textual and visual modalities, ensuring that retrieved content is contextually accurate and semantically aligned. The system is specifically optimized for scenarios involving extensive document understanding, leveraging pre-trained multi-modal mod els to process and retrieve information from structured and unstructured data sources (Cho et al., 2024).

3) **VisRag Approach:**

VisRAG employs a unified vision-language embedding space for retrieval, optimized for multi-modal document-level tasks. Its retrieval process integrates seamlessly into the generation pipeline, ensuring efficient and contextually relevant information retrieval. For a generation, VisRAG is tailored to synthesize information-rich responses by leveraging pre-trained transformers tuned for multi-modal data. The tightly coupled RAG system implemented by VisRAG ensures that retrieval and generation are closely integrated, focusing on document-level tasks to provide coherent and context-aware output.

C. Results

Metric	Qwen/Qwen2-VL-2B + ColPali	Qwen/Qwen2-VL-7B + Vidore/ColPali-v1.2	VisRAG	meta-llama/Llama-3.2-11B-Vision-Instruct
BLEU Score	Lower: 0.0485	Higher: 1.0000	Moderate: 0.3283	Higher: 1.0000
Inference Time	Slower: 3.2245 seconds	Faster: 0.0001 seconds	Equal: 0.0001 seconds	Moderate: 0.033959 seconds
Memory Usage (Generation)	Lower: CPU 5.91 GB, GPU 7.04 GB	Higher: CPU 6.23 GB, GPU 21.19 GB	Moderate: CPU 8.24 GB, GPU 13.37 GB	Moderate: CPU 5.60 GB, GPU 5.60 GB
Memory Usage (Retrieval)	Lower: CPU 5.91 GB, GPU 7.04 GB	Higher: CPU 6.23 GB, GPU 21.19 GB	Moderate: CPU 8.24 GB, GPU 13.37 GB	Moderate: CPU 5.60 GB, GPU 5.60 GB
Precision@3	Equal: 0.3333	Equal: 0.3333	Equal: 0.3333	Higher: 1.0000
ROUGE Scores	Lower: ROUGE-1: 0.1429, ROUGE-2: 0.0976, ROUGE-L: 0.1429	Higher: ROUGE-1: 1.0000, ROUGE-2: 1.0000, ROUGE-L: 1.0000	Moderate: ROUGE-1: 0.5589, ROUGE-2: 0.5510, ROUGE-L: 0.5534	Higher: ROUGE-1: 1.0000, ROUGE-2: 1.0000, ROUGE-L: 1.0000
Context Recall	Lower: 0.60	Higher: 1.00	Moderate: 0.70	Moderate: 0.50
Context Relevancy	Lower: 0.75	Higher: 0.83	Lower: 0.65	Moderate: 0.637039
Answer Similarity	Lower: 0.89	Higher: 0.92	Moderate: 0.66	Higher: 1.0000
Answer Correctness	Equal: 1.00	Equal: 1.00	Equal: 1.00	Equal: 1.00
Context Entity Recall	Higher: 1.00	Lower: 0.40	Higher: 1.00	Moderate: 0.50
Context Utilization	Equal: 1.00	Equal: 1.00	Lower: 0.99	Equal: 1.00

Fig. 7, Performance matrix

In Fig. 7, we fixed our data and we asked the same query to evaluate the performance matrix of the model's framework using a set of well-established evaluation metrics. These metrics measure various aspects of system performance, including text generation, retrieval accuracy, response quality, and memory usage. The goal is to assess how effectively both models handle multi-modal data (text and images) and provide relevant, contextually accurate responses to user queries. Query:" Tell me about Coffins of Padiamun 7."

VI. APPLICATIONS AND USE CASES

The AI-Enhanced Chatbot for Learning about Ancient Egypt offers a wide range of applications and use cases, addressing the needs of both educational institutions and tourism. By leveraging advanced AI technologies, this chatbot aims to enhance learning experiences and provide valuable insights into ancient Egyptian civilization.

A. Tourism

The chatbot is also designed to support tourists visiting historical sites and museums related to ancient Egypt. Its applications in the tourism sector include: Real-Time information: Tourists can interact with the chatbot to receive detailed information about historical sites, artifacts, and exhibits. By providing context and answering questions, the chatbot enhances the visitor experience, making it more informative and engaging.

- **Guided Tours:**

The chatbot can offer virtual guided tours, allowing tourists to explore ancient Egyptian sites through interactive lessons and multimedia content. This feature can be particularly beneficial for those unable to visit in person, providing a virtual experience that replicates on-site learning.

- **Cultural Context:**

The chatbot can provide insights into the cultural significance of various artifacts and sites, helping tourists understand the historical context of what they are observing. This enriches their experience and fosters a deeper appreciation for ancient Egyptian civilization.

B. Lifelong Learning

The chatbot serves as a valuable resource for lifelong learners interested in ancient Egypt it can support in:

Self-Directed Learning: Individuals seeking to expand their knowledge can engage with the chatbot to explore a wide range of topics at their own pace. The chatbot's quizzes and lessons provide opportunities for self-assessment and knowledge retention.

- **Community Engagement:**
By facilitating discussions and knowledge-sharing, the chatbot can foster a community of enthusiasts who share a passion for ancient Egyptian history. Users can participate in forums or groups, enhancing their learning experience through social interaction.

C. Research and Development

Researchers and historians can also benefit from the chatbot's capabilities:

- **Data Collection:**
The chatbot can gather user interactions and feedback, providing valuable insights into learning preferences and knowledge gaps. This data can inform future research and development in the field of educational technology.
- **Resource Accessibility:**
The chatbot can serve as a repository of knowledge, making research materials and historical documents easily accessible to scholars and students alike. This enhances academic research on ancient Egypt and encourages interdisciplinary collaboration.

VII. GAP ANALYSIS AND OPEN ISSUES

The AI-Enhanced Chatbot for Learning About Ancient Egypt is an exciting project with great potential, but it faces several gaps and challenges that must be addressed to ensure its success.

One significant issue is the lack of specialized datasets focused specifically on ancient Egyptian history. Much of the chatbot's information comes from online sources, which makes it challenging to guarantee accuracy and reliability. Building trust with users requires the chatbot to rely on highly credible sources like museum websites and academic research. This data must also be thoroughly validated, curated, and updated regularly to maintain its integrity over time.

Another gap lies in the seamless integration of multimodal features, such as combining text and images to create a rich and interactive experience. While these features enhance user engagement, aligning them naturally and cohesively is a complex task. Additionally, users will have varying levels of knowledge about ancient Egypt, ranging from beginners to experts. Most existing AI systems struggle to adapt their responses to fit the user's expertise or preferred learning style, which limits the chatbot's accessibility and effectiveness.

Ethical concerns present another important challenge. Ancient Egyptian history is deeply significant, and any errors, oversimplifications, or cultural insensitivity in the chatbot's responses could lead to misunderstandings or even offend users. To address this, the project requires guidance from historians and cultural experts to ensure the information presented is both accurate and respectful, safeguarding the cultural integrity of ancient Egyptian heritage.

Scalability is another critical issue. As new archaeological discoveries emerge, the chatbot's knowledge base must be continuously updated to remain relevant and accurate. This involves an ongoing effort to collect, integrate, and verify new data efficiently. Furthermore, the chatbot relies on LLAMA 3.2, a relatively new AI model with limited research or practical applications to guide its use. Customizing and fine-tuning this model for real-time, multimodal interactions adds another layer of complexity and requires extensive experimentation and iteration.

Overcoming these gaps will require a collaborative and well-rounded approach. By bringing together technical experts, historians, and cultural advisors, the project can create a chatbot that is not only intelligent and user-friendly but also culturally authentic and respectful. Addressing these challenges will enable the chatbot to become a groundbreaking tool for learning about and connecting with ancient Egyptian history.

VIII. FUTURE DIRECTION

The AI-Enhanced Chatbot for Learning about Ancient Egypt has great potential for further development and expansion, both in terms of functionality and reach. As the project evolves, several key directions will be pursued to enhance the user experience, broaden the application of the chatbot, and ensure it meets the needs of a global audience. These future directions aim to integrate cutting-edge technologies, expand content, and improve accessibility to create an even more immersive and interactive experience for users.

A. Integration with the Egyptian Museum

One of the primary goals for the future of this project is to integrate the AI chatbot into the infrastructure of the Egyptian Museum. This will ensure that all visitors, regardless of whether they have their own devices or not, can access the chatbot's capabilities to learn more about ancient Egyptian history, monuments, and artifacts during their visit.

Key Benefits:

- **Personalized Experience:**
Visitors can interact with the chatbot on their personal devices, receiving tailored information about the exhibits in real-time. This integration would provide context and deeper insights into the museum's displays, enhancing the visitor's experience.
- **Seamless User Interface:**
The chatbot can be designed to be easily accessible through a mobile app or museum-provided devices, ensuring that all visitors can engage with the chatbot with minimal friction.

- **Offline Accessibility:**
  Ensuring the chatbot works effectively in areas with limited internet access, such as parts of the museum, will be crucial to maximizing its utility.

B. Expansion of Content and Features

As the project grows, the expansion of content and features will be essential to ensuring that the chatbot remains informative, interactive, and engaging for users. A few key areas for content expansion include:

1) Interactive Storytelling:
The chatbot will incorporate interactive storytelling features that allow users to immerse themselves in key events from ancient Egyptian history. This feature will enable users to explore different perspectives, making choices that influence the outcome of historical events, thereby providing a more engaging and personalized experience.

Key Benefits:

- **User-driven Exploration:**
  Users can engage with historical events in a dynamic way, exploring various scenarios and outcomes based on their choices. This could involve role-playing elements, where users make decisions as historical figures or navigate complex events.
- **Enhanced Learning:**
  By experiencing historical events through interactive storytelling, users will develop a deeper understanding of the complexity and significance of those events in Egyptian history.

IX. CONCLUSION

The development of the AI-Enhanced Chatbot for Learning about Ancient Egypt represents a significant advancement in leveraging artificial intelligence and multimodal technologies to engage users with one of the world's most fascinating and ancient civilizations. By combining natural language processing, image retrieval, and real-time data interaction, this chatbot aims to provide users with a personalized, dynamic, and immersive learning experience, whether in the context of education or tourism.

However, creating such a system presents several challenges that must be carefully addressed. Ensuring data quality, cultural sensitivity, and the ethical integrity of the AI's interpretations of ancient Egyptian history are paramount. Additionally, the technical complexity of integrating multimodal content and real-time interactions requires the development of robust backend systems capable of handling diverse and dynamic user queries. The relatively uncharted territory of working with LLAMA 3.2 further adds to the complexity, requiring innovative solutions for customization, optimization, and scaling.

Despite these hurdles, the future of the chatbot holds great promise. With plans for integration into the Egyptian Museum and continuous efforts to expand its functionality, the project is set to enhance user experiences for both tourists and learners alike.

In conclusion, by addressing the identified challenges with careful planning, collaboration with experts, and iterative development, the AI-Enhanced Chatbot for Learning about Ancient Egypt has the potential to become a transformative tool for educating and engaging people with the ancient world. The project not only aims to enhance the visitor experience at the Egyptian Museum but also aspires to contribute to the broader field of AI-powered educational tools, pushing the boundaries of how technology can bring history to life.

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